

JEE MAIN REVISION MASTER: MOTION IN A STRAIGHT LINE 2026 2 QUESTIONS 3.33% WEIGHTAGE $\uparrow$ 99.4%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Motion In A Straight Line - Core Application	11 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Motion In A Straight Line - Core Application (High Yield — 11 PYQs)

Core Principle: Key problem pattern in Kcet Motion In A Straight Line.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The velocity of a particle moving along x -axis is given as $V=x^2-5x+4$ (in m/s) where x denotes the x -coordinate of the particle ..." [Ans: C]
- [2026 Shift] Q#2 (MCQ): "A car covers the first half of the distance between two places at 40 km/h and another half at 50 km/h. The average speed of the car is..." [Ans: B]
- [2025 Shift] Q#3 (MCQ): "Two stones begin to fall from rest from the same height, with the second stone starting to fall Δt seconds after the first falls..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The velocity of a particle moving along x-axis is given as $V=x^2-5x+4$ (in m/s) where x denotes the x-coordinate of the particle in metres. The magnitude of the acceleration of the particle when the velocity of the particle zero is

- (A) 2m/s^2
- (B) 3m/s^2
- (C) Zero **[Correct]**
- (D) 1m/s^2

Correct Answer: C

Solution Analysis: The velocity is given as a function of position: $v=x^2-5x+4$ We need acceleration when the velocity is zero. First find the positions where $v=0$: $x^2-5x+4=0$ Factorising, $(x-1)(x-4)=0$ So, $x=1\text{m}$ or $x=4\text{m}$ Now, since velocity is given as a function of x, acceleration is not simply (dv/dx) . We use $a=(dv/dt)$ Using chain rule, $a=(dv/dx) \cdot (dx/dt)$ But $(dx/dt)=v$ So, $a=v(dv/dx)$ Here, when the velocity is zero, $v=0$ Hence, $a=0 \cdot (dv/dx)=0$ So the magnitude of acceleration is 0m/s^2 Correct option: C